MATERIALS SCIENCE

Precise synthesis of targeted noble metal–based high-entropy alloy nanomaterials

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Noble metal–based high-entropy alloys (NM-HEAs) are promising catalysts because of their diverse compositions and complex surface structures, although understanding their formation and achieving precise atomic control remain challenging. Our team introduces a cost-effective synthesis method combining spray drying and thermal decomposition-reduction to produce HEA three-dimensional nanoframeworks (HEA-3DNFs) from quinary to octonary compositions, including Pt-, Ir-, Ru-, and PtRu-based HEAs. Mild heating in H2 facilitates simultaneous reduction of multiple elements, forming single-phase solid solution alloys. The PtNiCoCuRuIr HEA-3DNF exhibits superior methanol oxidation reaction performance (2637 milliamperes per milligram Pt), 15 times higher than commercial Pt black. Noble metals (Pt, Ru, and Ir) lower activation energy for reducing transition metals (Ni, Co, Cu, and Fe). In situ scanning transmission electron microscopy and density function theory calculations reveal phase transformation from a Pt-based face-centered cubic to Ru-based hexagonal close-packed structure, with control over phase composition by adjusting reduction temperatures and Pt-to-Ru ratios. These findings offer insights for developing high-performance and tunable NM-HEA catalysts.

INTRODUCTION

High-entropy alloy (HEA) nanomaterials, comprising a minimum of five distinct elements, demonstrate both thermodynamic and kinetic stability propelled by a high-entropy effect. Their diverse composition and intricate surface atomic structures make them particularly adept at showcasing exceptional catalytic performance (*1*–*9*). Moreover, current research indicates that nearly all reported HEA nanomaterials incorporate noble metal elements. Given the widespread use of noble metals and their alloys in electrocatalysis, the emergence of noble metal–based HEA (NM-HEA) nanomaterials holds a pivotal role in future applications (*3*–*24*). These materials exhibit remarkable performance and substantial potential across various catalytic domains, leveraging the advantageous properties of both NMs and HEA (*25*). Their applications span a wide spectrum, including NH_3 decomposition (*4*), NH_3 oxidation (*5*), electrolytic water hydrogen evolution (*7*, *10*, *11*), oxygen evolution and reduction (*12*–*17*), CO_2/CO reduction (*18*, *19*), and methanol and ethanol oxidation (*20*–*23*). Nowadays, there are already many ways to synthesize NM-HEA nanomaterials, such as the carbothermal shock method, fast moving bed pyrolysis strategy, microwave heating method, polyol method, dealloying method, etc. (*26*, *27*), but devising a straightforward, cost-effective, and scalable synthesis route for NM-HEA nanomaterials remains a persistent challenge. Moreover, comprehending the formation mechanism of NM-HEA nanomaterials, especially the role played by NM elements, holds paramount importance in tailoring their composition and structure. Notably, there is a paucity of research in this area, particularly regarding NM. Previous studies in our group have demonstrated the crucial role of platinum in the formation of Pt-based HEA nanomaterials, showing that the addition of Pt facilitated the creation of a single-phase solid solution [face-centered cubic (fcc)] when combined with other metals (*23*, *24*). In contrast, systems lacking Pt, such as CoNiFe and CoCuNiFe nanoparticles, exhibited phase separation, element segregation, uncontrolled micromorphology, and high thermal decomposition temperatures. Hence, investigating the fundamental mechanism of NM elements (Pt, Ru, Ir, Pd, etc.) in the formation process of NM-HEA nanomaterials holds immense great meaning and presents a challenging frontier.

Besides, the precise manipulation of crystal structures at the atomic level within single nanoparticles is of great importance in expanding application spaces and enhancing performance, because diverse crystal structures formed by the same elements may react at different rates in identical reactions because of the varying binding energy across distinct crystal surfaces (*28*–*31*). Despite this significance, factors influencing the crystal structure of alloys remained scarcely explored, primarily because controlling the crystal structure of an alloy with an identical composition poses an exceptionally challenging task (*32*–*35*). While numerous studies have delved into single-phase solid solution in NM-HEA nanoparticles, a rational strategy to selectively control their crystal structures is not reported and well developed yet (*3*, *8*, *36*). This challenge is amplified in NM-HEA because of the complexity arising from multiple interacting elements (more than five) and intricate synthesis conditions involving rapid heating and cooling rates. Unlike simpler alloys with fewer elements, the complexity of NM-HEA makes achieving crystal structure control notably more demanding. Moreover, it is deemed that a higher count of atoms with different crystal structures exposed on the surface within the same nanoparticle enhances catalytic reactions (*2*, *3*, *6*, *25*–*27*). Consequently, investigating methods to precisely design and control specific crystal structures, while maximizing the exposure of surface

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atoms, becomes inevitable for the advancement and application of NM-HEA nanomaterials.

Herein, we use a straightforward, cost-effective, and universal synthesis method—the spray drying technique in conjunction with thermal decomposition-reduction (SD-TDR)—to synthesize NM-HEA nanomaterials, characterized by their three-dimensional nanoframework (3DNF) structure. In this work, our team successfully synthesized multiple batches of NM-HEA-3DNFs, including PtNiCoCuRu, PtNiCoCuFe, PtNiCoCuRuIr, PtNiCoCuRuIrPd, PtNiCoCuRuIrPdFe, IrNiCoCuFe, and RuNiCoCuFe, highlighting the versatility of the SD-TDR approach. These NM-HEA-3DNFs showcased a highly porous and self-supported architecture, assembled through staggered nanoarms serving as fundamental building blocks. In addition, the methanol oxidation reaction (MOR) was investigated on the as-synthesized Pt-based HEA-3DNFs, which presented superior electrocatalytic performance. Moreover, our study delved into the role of NM elements (Pt, Ru, Ir, and Pd) in the formation mechanism of NM-HEA-3DNFs using both experimental and theoretical approaches. Furthermore, a presentative sample of $Pt_xNiCoCuRu_y$ with changing compositions (x:y = 3:1, 2:1, 1:1, 1:2, 1:3, and 1:4) was synthesized at various reduction temperatures to explore the phase transition mechanism from fcc to hexagonal close-packed (hcp) in NM-HEA-3DNFs by in situ scanning transmission electron microscopy (STEM) experiments and density function theory (DFT) calculations. In principle, these results successfully achieve the desired relative content of target phase structures (fcc and/or hcp) in PtNiCoCuRu HEA-3DNFs, marking a substantial milestone in the pioneering control of dual-phase structures within a single NM-HEA nanostructure.

RESULTS

Rational design of NM-HEA-3DNFs

Figure 1A presents the synthetic process of NM-HEA-3DNFs containing two key steps. The initial step is spray drying, which is essential for achieving a homogeneous mix of multiple elements within the precursor particles. In this step, $(NH_4)_2PtCl_6$, M salts (M = Co, Cu, Ni, Fe, Ru, Ir, and Pd), and NH_4Cl were dissolved in deionized water to form a precursor solution, which was then spray dried using a custom-designed spray dryer to produce a well-mixed

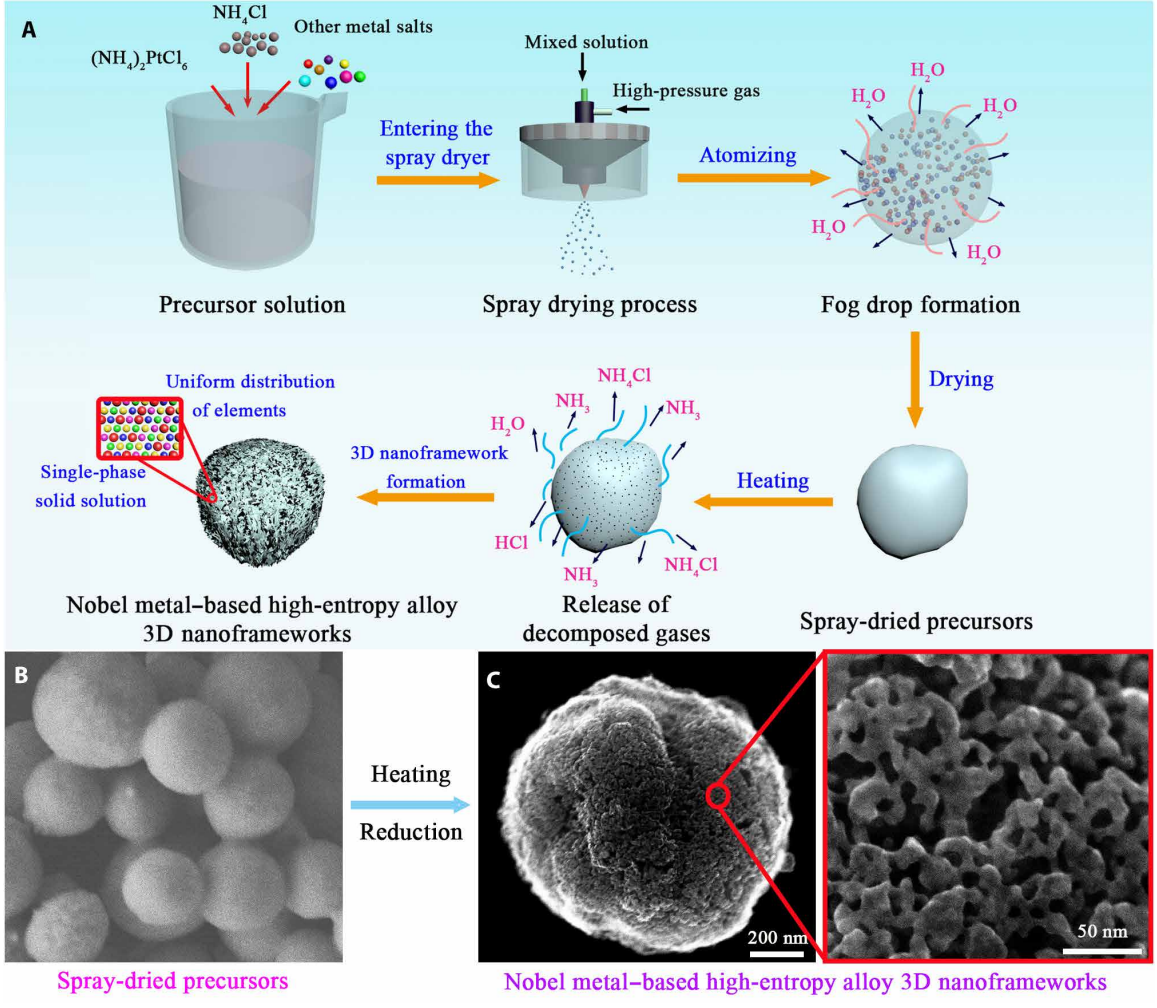


Fig. 1. Schematic diagram of the synthetic process and microscopic properties of the Pt-based HEA-3DNF. (**A**) Synthetic process of NM-HEA-3DNF. (**B**) SEM image of spray-dried well-mixed precursors. (**C**) Second electron STEM images of NM-HEA-3DNF.

precursor. Figure 1B and fig. S1 (see details in the Supplementary Materials) further elucidate that the morphology of the spray-dried precursors is near spherical and the elements are evenly distributed, as examined by scanning electron microscopy (SEM) imaging and energy-dispersive x-ray spectroscopy (EDS) elemental mapping. The second essential step involves the thermal decomposition-reduction (TDR), which supports the process leading to the formation of NM-HEA-3DNFs. During the TDR process, the spray-dried precursors were reduced in the presence of a H_2 atmosphere, with a moderate heating rate of 3°C/min, a relatively low thermal decomposition temperature set at 300°C, a duration time of 30 min, and a controlled cooling rate of 5° to 10°C/min. Subsequently, the eight immiscible metallic elements, each with distinct atomic characteristics as listed in table S1, were alloyed within the nanostructure to form a single-phase solid solution. Simultaneously, during the decomposition stage, the release of gases like HCl, NH_4Cl, NH_3, and H_2O created a substantial number of channels, forming a porous structure as these gases escaped from the precursors. Ultimately, this process led to the formation of NM-HEA-3DNFs shown in Fig. 1C. Here, the Pt-based (PtNiCoCuRu, PtNiCoCuRuIr, PtNiCoCuRuIrPd, and PtNiCoCuRuIrPdFe) HEA-3DNFs were successfully synthesized. The SEM and second electron SEM images in fig. S2 display the quasi-spherical 3DNF structure in the microscale. The STEM-EDS elemental mappings containing PtNiCoCuRu, PtNiCoCuRuIr, and PtNiCoCuRuIrPd HEA-3DNFs in Fig. 2 (A to C) indicate the homogeneous elemental distribution without obvious agglomeration or segregation. Meanwhile, these as-formed HEA-3DNFs were single-phase solid solutions with an fcc structure, without a secondary phase or intermetallic compounds, as confirmed by the x-ray diffraction (XRD) profiles in Fig. 2D.

In particular, Fig. 2 (E to I) further analyzes the octonary PtNiCoCuRuIrPdFe HEA-3DNF by the high-angle annular dark-field STEM (HAADF-STEM) and annular bright-field (ABF-STEM) imaging methods. Figure 2 (E to G) confirms that the PtNiCoCuRuIrPdFe HEA-3DNF has a highly porous and self-supported structure, assembled from staggered nanoarms. Moreover, the atomic-scale HAADF-STEM images and fast Fourier transform (FFT) in Fig. 2H and fig. S3A provide additional support of the atomistic arrangement of the fcc structure from both <011> and <−112> orientations. The uniform distribution of all elements is also observed through STEM-EDS elemental mapping, as shown in Fig. 2I and fig. S3B. Furthermore, the microcosmic and macroscopic compositions were in good agreement with the original design of the alloy component, proved by STEM-EDS spectroscopy (fig. S4) and inductively coupled plasma mass spectroscopy (table S2). The fully reduced PtNiCoCuRuIrPdFe sample (with no chlorine content) and the bonding state of metals were also confirmed by x-ray photoelectron spectroscopy (XPS), as shown in fig. S5. Thus, Pt-based HEA-3DNFs with single-phase solid solution were successfully achieved using the SD-TDR method.

Electrocatalytic performance of HEA-3DNFs in MOR

After the analysis of the microstructure and elemental composition of the above HEA-3DNF, a presentative sample of PtNiCoCuRuIr HEA-3DNF was selected to perform MOR as anodic electrocatalysts for direct methanol fuel cells. The MOR performance was tested in an aqueous solution of 0.5 M H_2SO_4 and 0.5 M CH_3OH, and a commercial catalyst [Pt black, Johnson Matthey (JM)] was selected as the control group. To ensure the effectiveness of the MOR performance, we tested three PtNiCoCuRuIr HEA-3DNF samples prepared under the same conditions. The electrochemically active surface area was calculated from the hydrogen underpotential desorption region in fig. S6. The electrochemically active surface area of PtNiCoCuRuIr HEA-3DNF was 104.4 m^2/g_{Pt} (equivalent to 41.7 m^2/g_{PtIrRu}), which was much larger than that of JM-Pt black (17.59 m^2/g_{Pt}). The cyclic voltammetry (CV) curves of MOR were normalized by the mass of Pt and the combined mass of Pt, Ir, and Ru, as illustrated in Fig. 3A and fig. S7A. The sample of PtNiCoCuRuIr HEA-3DNF exhibited superior electrocatalytic activity, reflected in its peak mass current density, whether normalized by Pt mass (mA/mg_{Pt}) or PtIrRu mass (mA/mg_{PtIrRu}). The MOR mass activity of PtNiCoCuRuIr HEA-3DNF reached 2637 mA/mg_{Pt} (equivalent to 1054.23 mA/mg_{PtIrRu}), which was ~15 times (or 6 times) higher than that of JM-Pt black (170 mA/mg_{Pt}), as depicted in Fig. 3B and figs. S7B and S8. Furthermore, the chronoamperometry (CA) measurements were carried out to evaluate the stability of catalysts. Figure 3C shows the CA curves, and the current density of PtNiCoCuRuIr HEA-3DNF is much higher than that of JM-Pt black during the entire time range. Therefore, the PtNiCoCuRuIr HEA-3DNF catalysts present superior MOR electrocatalytic performance over the commercial JM-Pt black catalyst, confirming that NM-HEA-3DNF is a kind of advanced catalyst with promising application foreground.

Formation mechanism of single-phase solid solution in NM-HEA-3DNF

Given the superior electrocatalytic performance of NM-HEA-3DNF, identifying the key elements and understanding the formation mechanism of the single-phase solid solution in NM-HEA-3DNFs appear to be particularly substantial. Typically, the individual elements (Pt, Ni, Co, Cu, Fe, Ru, Ir, and Pd) have distinct reduction and decomposition temperatures, making simultaneous reduction to individual metals at lower temperatures (300°C) unfeasible (*4*, *23*, *24*). However, in the synthetic process, these elements were concurrently reduced to form a single-phase solid solution at 300°C in the presence of a H_2 atmosphere. This unexpected outcome led us to conduct an in-depth investigation into the underlying mechanism. The results showed that Pt, Ru, Ir, and Pd salts were able to be reduced at 300°C in H_2, while the salts of the nonprecious metals (Ni, Co, Cu, and Fe) still existed in salt form or oxide, as examined in the XRD spectra in fig. S9. Thus, it was speculated that NM elements may play an important role in the formation of HEA nanomaterials. Herein, we first chose the Pt element to study it in depth. In the absence of Pt, the spray-dried NiCoCuFe precursors remained in a salt form after being reduced at 300°C in H_2, as evidenced by the XRD pattern in Fig. 3D, whereas the precursors were reduced to a metal alloy when the temperature reached 450°C, and phase separation appeared as depicted by the XRD pattern in Fig. 3D and the STEM-EDS mappings in Fig. 3E. This phase separation led to the formation of distinct Ni-, NiCo-, and FeCo-rich areas, contributing to rapid nanoparticle growth and the destruction of the 3DNF (fig. S10). Consequently, the spray-dried NiCoCuFe precursors proved incapable of being reduced in H_2 at 300°C, resulting in phase-separated alloys instead of a single-phase solid solution, as illustrated in Fig. 3F. Moreover, it was found that Pt salts could be reduced to Pt metal in H_2, even at a low temperature of 200°C, as depicted in Fig. 3G. The SEM images of PtNiCoCuFe nanomaterials are displayed in fig. S11 (A and B), clearly revealing the formation of the 3DNF structure. Consequently, the addition of Pt to the NiCoCuFe system lowered

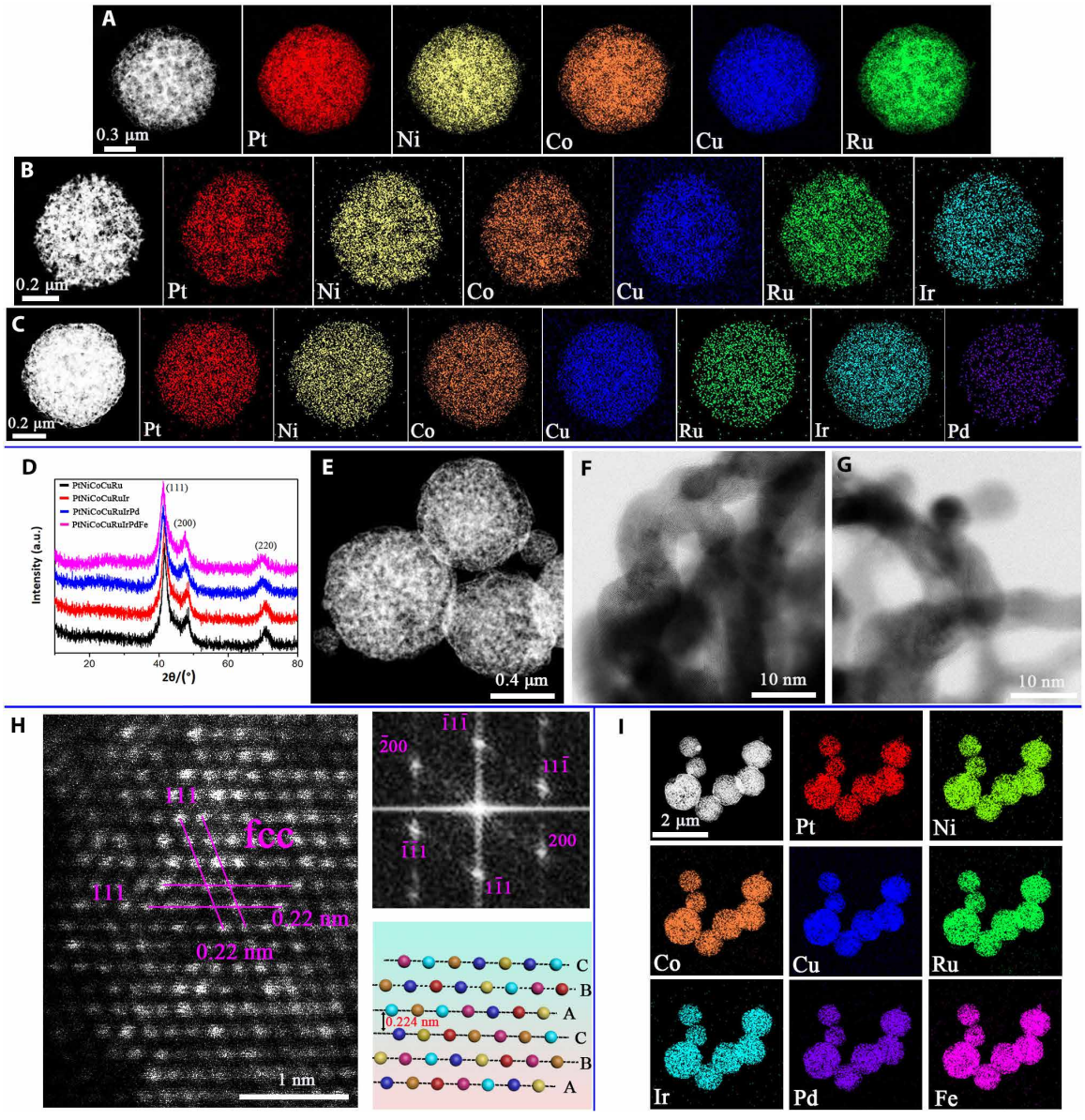


Fig. 2. Microscopic analysis of NM-HEA-3DNF. (**A** to **D**) STEM-EDS elemental mappings (A to C) and XRD profiles (D) of PtNiCoCuRu, PtNiCoCuRuIr, PtNiCoCuRuIrPd, and PtNiCoCuRuIrPdFe NM-HEA-3DNFs. a.u., arbitrary units. (**E** to **I**) High-magnification HAADF-STEM (E), ABF-STEM (F and G), and atomic-scale HAADF-STEM images, corresponding FFT patterns (H), and STEM-EDS elemental mappings (I) of PtNiCoCuRuIrPdFe HEA-3DNF. The inset in (H) shows the atomistic arrangement model of the fcc structure.

the reduction temperature of Ni, Co, Cu, and Fe, promoting the formation of PtNiCoCuFe HEA-3DNF, as confirmed by Fig. 3H and fig. S11C and illustrated in Fig. 3I. Furthermore, to further examine the effect of Pt, we synthesized a series sample of PtNi, PtCo, PtCu, and PtFe, showing that the spray-dried PtNi, PtCo, PtCu, and PtFe precursors were reduced to single-phase alloy at 300°C in H_2, as presented in fig. S12. Therefore, it is further proved that the Pt element can reduce the reduction temperature of Ni, Co, Cu, and Fe and facilitate the formation of single-phase solid solution alloy.

On the basis of the above phenomenon and the discussions in the previous reports, it was speculated that H_2 may react with the newborn Pt atom on the surface to facilitate the formation of single-phase solid solution alloy (*37*–*39*). To study this, we further performed the DFT calculation. The interaction of H_2 with the newborn Pt metal on the surface of PtNiCoCuFe mixed precursors was dissected into three distinct reaction stages: H_2 adsorption, H_2 dissociation forming the H–Pt bond, and H desorption leading to highly reactive H atoms, as depicted in fig. S13. For an fcc crystal structure of Pt-based HEA-3DNF, the (111) crystal plane was consistently the most prominent surface facet because of the lowest surface energy, which was also confirmed by XRD patterns in Fig. 2D and atomic-scale HADDF-STEM image in Fig. 2H. Therefore, the free energy profiles for the stages of H_2 adsorption, H_2 dissociation, and H desorption on the Pt(111) crystal plane, as shown in

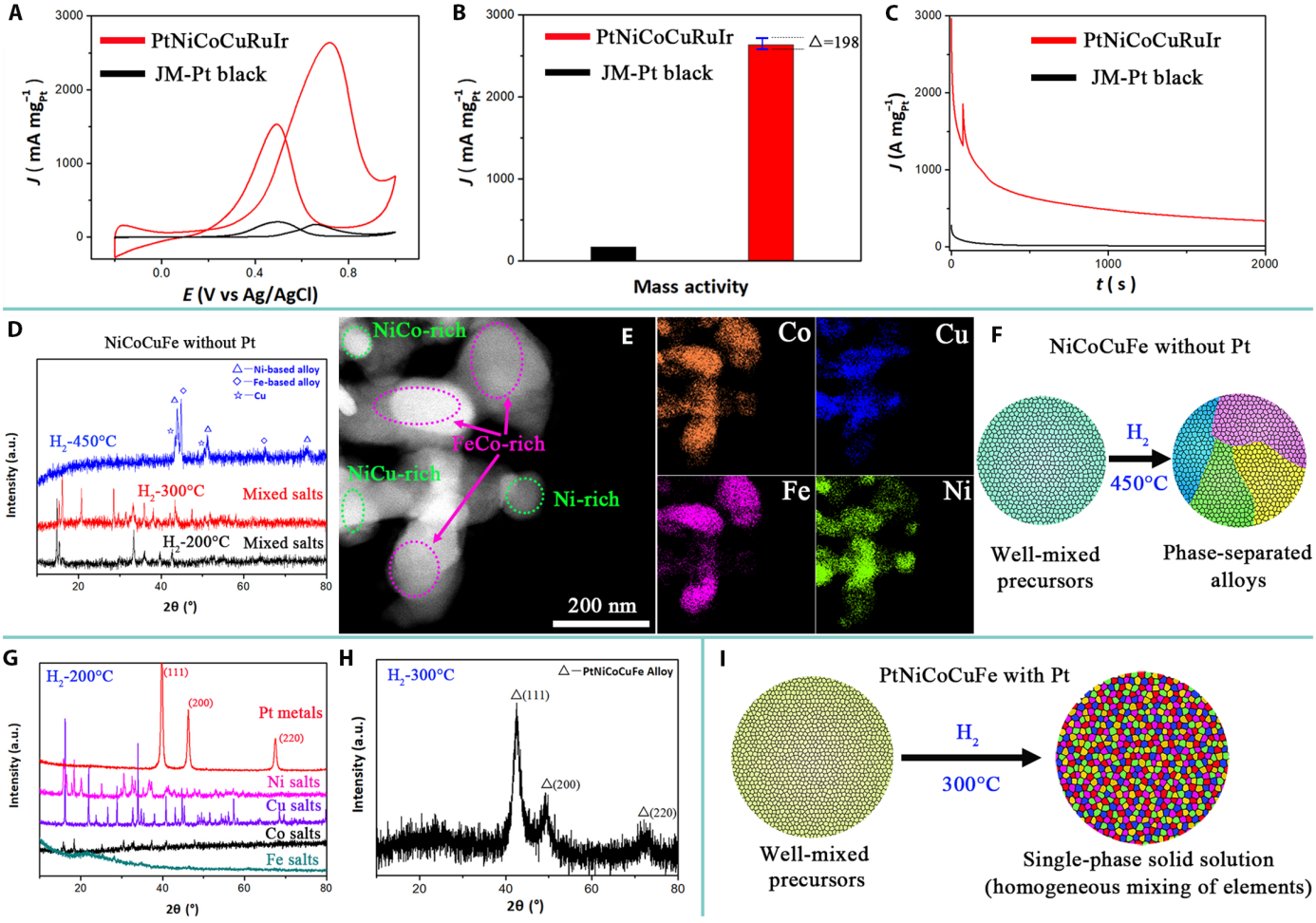


Fig. 3. Electrocatalytic performance and formation mechanism of NM-HEA-3DNF. (**A**) CV profiles (mass peak current) of PtNiCoCuRuIr HEA-3DNF and Pt black (JM) for the MOR tested in an aqueous solution of 0.5 M H_2SO_4 and 0.5 M CH_3OH with a scan rate of 50 mV/s. (**B**) Bar graph illustrating the current density of mass activity at peak potential. (**C**) CA curves of PtNiCoCuRuIr HEA-3DNF and Pt black (JM) for the stability of MOR. (**D**) XRD spectra of NiCoCuFe without Pt salts obtained at reduction temperatures of 200°, 300°, and 450°C, respectively. (**E**) STEM-EDS elemental mappings of NiCoCuFe alloy obtained at 450°C. (**F**) Formation diagram of phase-separated alloy in the absence of Pt. (**G**) XRD spectra of Pt, Ni, Co, Cu, and Fe salts at a reduction temperature of 200°C. (**H**) XRD spectrum of PtNiCoCuFe HEA at a reduction temperature of 300°C. (**I**) Formation diagram of single-phase solid solution HEA in the presence of Pt.

Fig. 4A, were calculated on the basis of DFT calculations. The values of free energy for these stages were −0.643, −0.695, and −0.984 eV, respectively, all showing a downhill trend. The changes of the free energy values for H_2 adsorption ($\Delta G_{H2\text{-}ads}$), H_2 dissociation ($\Delta G_{H2\text{-}diss}$), and H desorption ($\Delta G_{H\text{-}des}$) were −0.643, −0.052, and −0.289 eV respectively, indicating that the reaction can spontaneously generate highly reactive H atoms. It was believed that these highly reactive H atoms effectively assisted the reduction of the other elements (Ni, Co, Cu, and Fe) nearby Pt to their metallic state. Subsequently, the newly formed Ni, Co, Cu, and Fe atoms simultaneously entered the crystal lattice of Pt, culminating in the creation of Pt-based HEA with a single-phase solid solution with an fcc crystal structure, as illustrated in Fig. 4B. Furthermore, the consumption of highly reactive H atoms during the reduction process of Ni, Co, Cu, and Fe promotes their continuous generation.

To validate the accuracy of the Pt-based HEA formation mechanism, we conducted the comparative assessments involving other NM elements, such as Ir, Ru, and Pd. Similarly, the XRD profiles in fig. S14 indicate that Ir, Ru, and Pd salts are also able to be reduced to Ir, Ru, and Pd metals at 200°C in H_2. DFT calculations were executed for H_2 adsorption, H_2 dissociation, and H desorption on the Ir(111) crystal plane and yielded free energy values of −0.516, −0.537, and −0.842 eV, respectively, as shown in Fig. 4C. These values indicate a downhill energy trend, similar to what was observed for the Pt metal. Consequently, it was deduced that Ir can form HEA nanomaterials alongside Ni, Co, Cu, and Fe elements. Therefore, the sample of IrNiCoCuFe was synthesized under identical conditions to examine this deduction. As expected, the IrNiCoCuFe HEA-3DNF with a single-phase solid solution (fcc) and uniform element distribution was achieved as confirmed by the XRD profiles, SEM images, HAADF image, and STEM-EDS mappings in Fig. 4D and fig. S15. Further DFT calculations on the Ru(011) crystal plane, as presented in Fig. 4E, revealed free energy values of −1.286, −1.326, and −2.197 eV for H_2 adsorption, H_2 dissociation, and H desorption, respectively. Subsequently, the RuNiCoCuFe sample synthesized under identical conditions demonstrated a single-phase solid

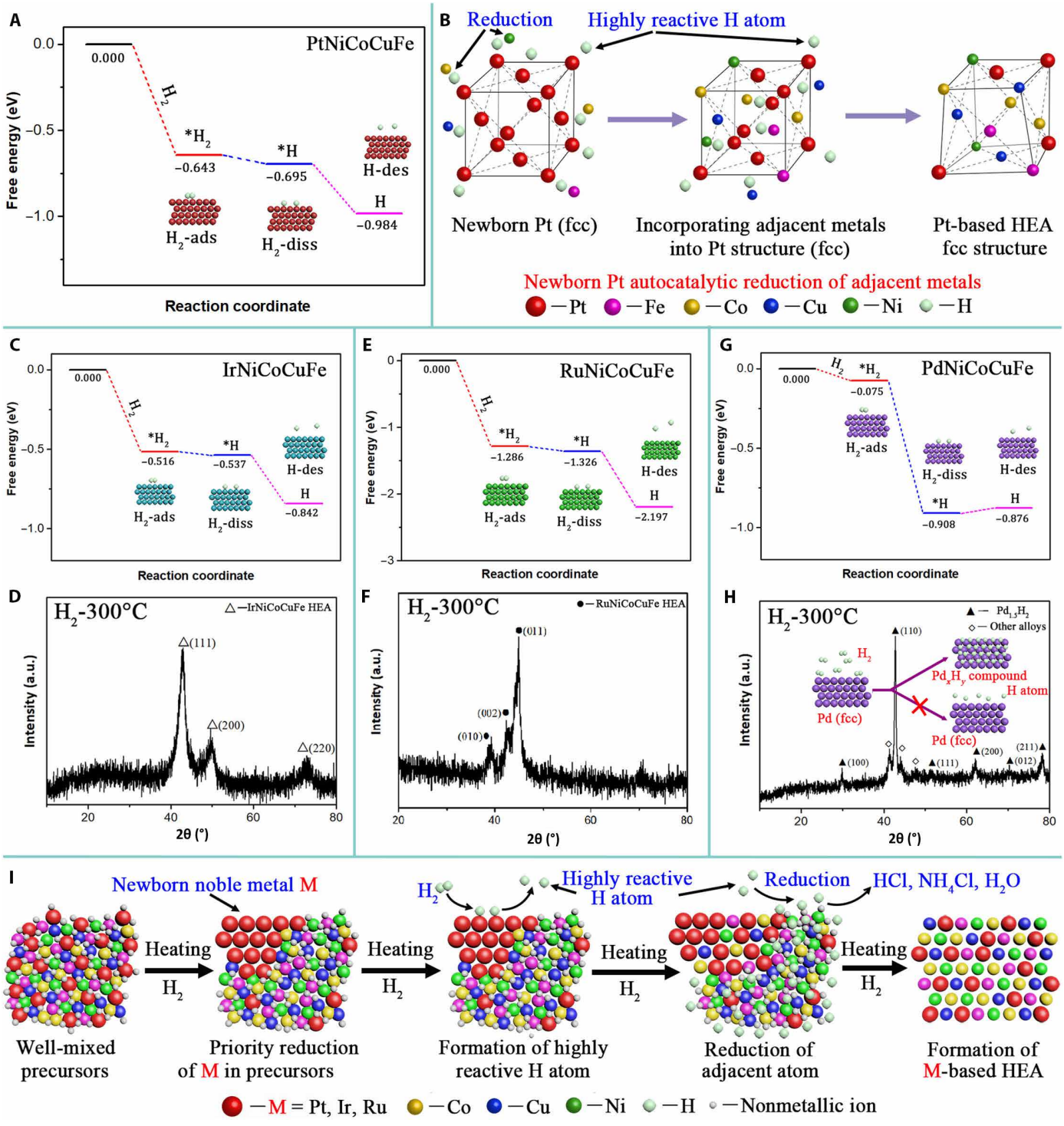


Fig. 4. Pinpointing key elements and formation mechanism of single-phase solid solution in NM-HEA-3DNF. (**A**) Gibbs free energy diagram of H_2 on the surface of the Pt(111) crystal plane. (**B**) Schematic diagram of newborn Pt autocatalytic reduction of adjacent metals by highly reactive H atoms. (**C**) Gibbs free energy diagram of H_2 on the surface of the Ir(111) crystal plane. (**D**) XRD spectrum of IrNiCoCuFe HEA at a reduction temperature of 300°C. (**E**) Gibbs free energy diagram of H_2 on the surface of the Ru(011) crystal plane. (**F**) XRD spectrum of RuNiCoCuFe HEA at a reduction temperature of 300°C. (**G**) Gibbs free energy diagram of H_2 on the surface of the Pd(111) crystal plane. (**H**) XRD spectrum of PdNiCoCuFe alloy at a reduction temperature of 300°C and formation diagram of the Pd_xH_y compound without highly reactive H atoms. (**I**) Description scheme of the formation mechanism of NM-HEA-3DNF.

solution (hcp), uniform element distribution, and a distinct 3DNF structure, which were confirmed through XRD profiles, STEM-EDS elemental maps, HAADF-STEM image, FFT, and SEM images depicted in Fig. 4F and fig. S16. It was important to emphasize that the hcp structure observed in the RuNiCoCuFe HEA originated from the Ru atoms, which, upon reduction, naturally adopt an hcp crystal structure. The subsequent reduction of Ni, Co, Cu, and Fe by highly reactive H atoms facilitated their incorporation into the existing hcp lattice, leading to the formation of a Ru-based HEA with an hcp crystal structure. These findings reinforce the proposed mechanism for HEA formation, highlighting the critical roles of NM in stabilizing phase structures.

Furthermore, DFT calculations were conducted on the Pd(111) crystal plane, with the corresponding free energy profiles illustrated in Fig. 4G. The values for H_2 adsorption, H_2 dissociation, and H desorption were −0.075, −0.908, and −0.876 eV, respectively, while the values for $\Delta G_{H2\text{-}ads}$, $\Delta G_{H2\text{-}diss}$, and $\Delta G_{H\text{-}des}$ were −0.075, −0.833, and 0.031 eV. Notably, the H-desorption free energy profiles of Pd(111) showed an uphill trend, contrasting with the downhill profiles observed for Pt, Ir, and Ru metals. The positive $\Delta G_{H\text{-}des}$ value on the Pd(111) surface indicates that the H atom is more difficult to desorb, making Pd less favorable for this process. Consequently, it was inferred that the Pd element went against the formation of HEA alongside Ni, Co, Cu, and Fe elements. This deduction was confirmed by the synthesis of the PdNiCoCuFe sample, wherein a Pd-based HEA was not obtained as evidenced by the XRD image in Fig. 4H. The microstructure observation and elemental distribution mappings (fig. S17) revealed distinct phase separations, such as Pd- and Ni-rich regions, aligning precisely with computational predictions. The elemental segregation and multiphase alloys in the PdNiCoCuFe sample also lead to the destruction of the 3DNF. In addition, the reaction energy barriers for H desorption on Pt(111), Ir(111), Ru(011), and Pd(111) crystal planes are evaluated in fig. S18. The energy barriers were found to be 0.27 eV for Pt(111), 0.31 eV for Ir(111), 0.29 eV for Ru(011), and notably higher at 0.65 eV for Pd(111). These results further indicate that H atoms are more likely to desorb from the surfaces of Pt, Ir, and Ru metals compared to Pd.

Therefore, the above comprehensive exploration involving Ru, Ir, and Pd elements convincingly validated the rationality behind the NM-HEA formation mechanism. This investigation identifies the key metal elements (M = Pt, Ir, and Ru) among the others, providing a detailed understanding of the formation mechanism, as illustrated in Fig. 4I. Initially, at temperatures below 300°C, the key M atoms within MNiCoCuFe mixed precursors were preferentially reduced, giving rise to the formation of an M-based crystal structure. Subsequently, H_2 underwent catalytic cleavage on the surface of newly formed M-based crystals, producing highly reactive H atoms. These highly reactive H atoms then facilitated the reduction of adjacent Ni, Co, Cu, and Fe elements near M atoms within the MNiCoCuFe mixed precursors to their metallic state. Ultimately, the MNiCoCuFe HEA was achieved, featuring a single-phase solid solution by integrating the newly formed Ni, Co, Cu, and Fe atoms into the M crystal structure.

Precise control on the atomic arrangements of $Pt_xNiCoCuRu_y$ HEA-3DNFs

The above findings highlighted the distinction between Pt-based (PtNiCoCuFe) and Ru-based (RuNiCoCuFe) HEAs in terms of their crystal structures, namely fcc and hcp configurations. This raised a crucial question: Was it plausible to precisely tailor the phase composition containing both fcc and hcp structures within NM-HEA nanostructures? This query represents an uncharted territory in previous research, yet it holds substantial potential across various catalytic scenarios. By providing a range of unique crystal surfaces and complex active sites, it opens avenues for enhancing catalytic performance through synergistic effects (*22*, *40*–*42*). Hence, we devised a strategy to control the phase composition of NM-HEA-3DNF, achieving targeted atomic arrangement from fcc to hcp structures by manipulating the ratio of Pt to Ru, along with the varied treatment temperatures, as depicted in Fig. 5A. Given the situation of $Pt_1Ni_1Co_1Cu_1Ru_1$ obtained at 300°C in H_2 exhibiting an fcc structure, as evidenced in the XRD profile (Fig. 2D), the atomic ratio of Pt to Ru was systematically altered across several samples—ranging from 3:1 to 2:1, 1:2, 1:3, and 1:4—designated as $Pt_3Ni_1Co_1Cu_1Ru_1$, $Pt_2Ni_1Co_1Cu_1Ru_1$, $Pt_1Ni_1Co_1Cu_1Ru_2$, $Pt_1Ni_1Co_1Cu_1Ru_3$, and $Pt_1Ni_1Co_1Cu_1Ru_4$, respectively. The XRD profiles of these $Pt_xNiCoCuRu_y$ HEA-3DNF samples were compared, as shown in fig. S19. Notably, the characteristic peaks of the hcp crystal structure emerged when the proportion of Ru exceeded that of Pt. This phenomenon stemmed from the relatively easier reduction of Pt in larger quantities compared to Ru, particularly at lower temperatures (fig. S20). Subsequently, the SEM images, STEM images, and STEM-EDS elemental maps of $Pt_1Ni_1Co_1Cu_1Ru_4$ (fig. S21) further validated the 3DNF structure and homogeneous distribution of elements in the $Pt_1Ni_1Co_1Cu_1Ru_4$ sample. Subsequently, both fcc and hcp structures were distinctly identifiable through the XRD spectrum (fig. S19F) and HAADF-STEM image with the corresponding FFT patterns, as shown in Fig. 5 (B to E). Moreover, Fig. 5 (F and G) further demonstrates the coexistence of these structures, with the fcc phase displaying an ABCABC… stacking sequence and the hcp phase showing an ABAB… sequence. Despite the high Ru content, the hcp structure does not dominate over the fcc structure, as confirmed by the XRD profile (fig. S19). This indicates that merely increasing Ru content is insufficient to control the phase transition from fcc to hcp in $Pt_xNiCoCuRu_y$ HEA-3DNF.

Moreover, the precise control over the phase composition within $Pt_xNiCoCuRu_y$ HEA was investigated by analyzing its cohesive energy (E_{coh}) through DFT calculations. It showed the optimized crystal structures of Pt-based (fcc) and Ru-based (hcp) HEA configurations with an evenly elemental distribution by replacing Pt and Ru sites with other atoms (Pt, Ni, Co, Cu, and Ru) in their respective crystal structures in fig. S22, and the atomic ratio is 1:1:1:1:1. The E_{coh} values for the optimized Pt-fcc HEA and Ru-hcp HEA were −7.946 and −12.833 eV, respectively. As shown in Fig. 6A, this result implied that the generation of Ru-hcp HEA from the free atomic state necessitated more energy release. The E_{coh} values from the Pt-fcc HEA structure to the Ru-hcp HEA structure is negative (−4.887 eV), as shown in fig. S23, indicating that Ru-hcp HEA has a more stable structure compared with the Pt-fcc HEA structure thermodynamically. However, on the basis of our experimental results, the PtNiCoCuRu HEA structure predominantly adopts the fcc structure during the initial reaction stages rather than the more thermodynamically stable hcp structure. To figure out this contradiction, the ΔG from the Pt-fcc HEA structure to the Ru-hcp HEA structure was calculated to study the spontaneity of this reaction, as depicted in Fig. 6B. The ΔG was −5.32 eV, indicative of a favorable tendency for spontaneous transformation from Pt-fcc to Ru-hcp HEA. Moreover, the dynamic energy barrier [activation energy (E_a)] is also evaluated in Fig. 6C, revealing a small energy barrier of 0.45 eV from the Pt-fcc structure

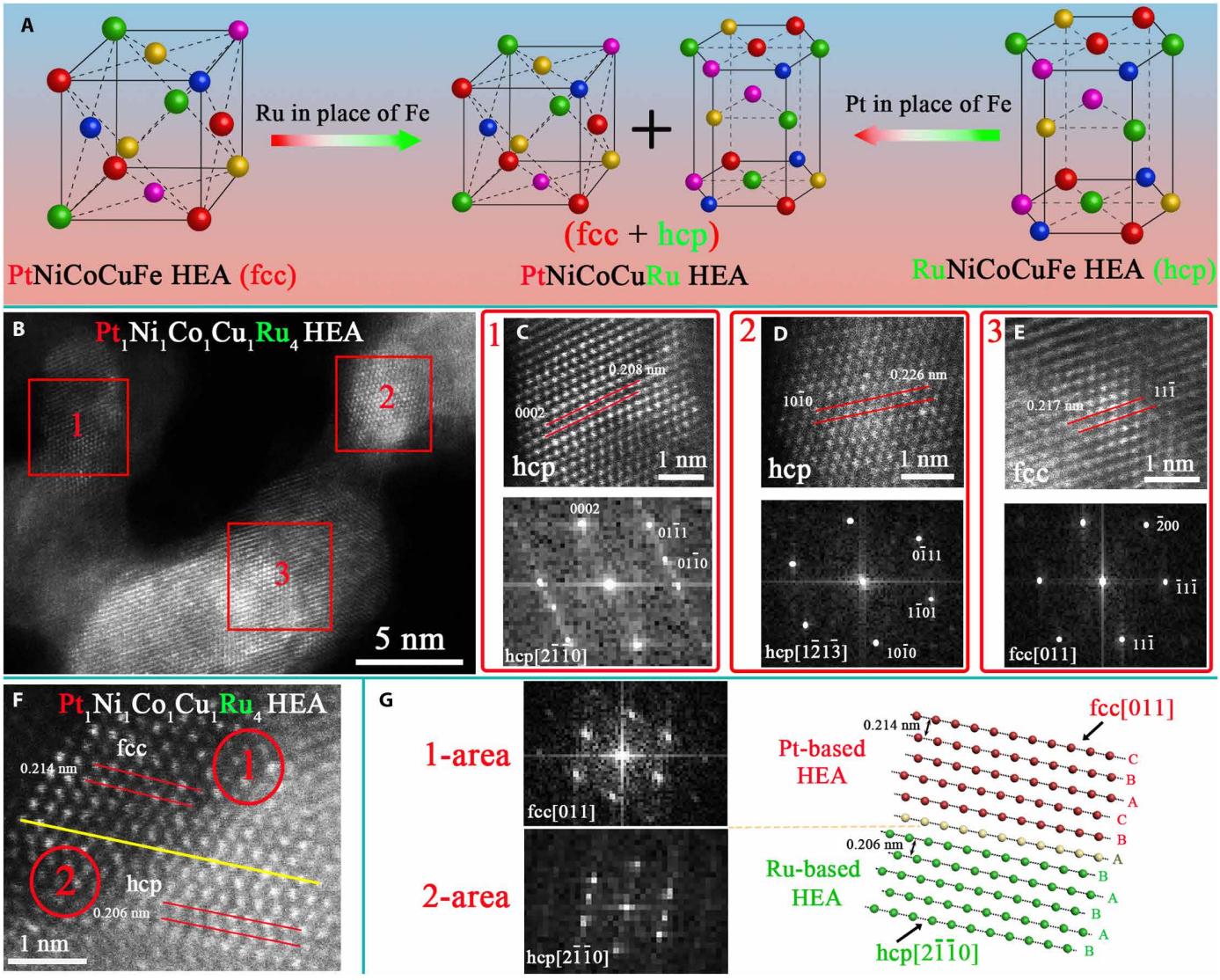


Fig. 5. Precise design of phase composition in PtNiCoCuRu HEA. (**A**) The illustration of PtNiCoCuRu HEA contains both fcc and hcp crystal structures. (**B**) HAADF-STEM images of the $Pt_1Ni_1Co_1Cu_1Ru_4$ HEA-3DNFs obtained at 300°C. The insets (**C** to **E**) are the corresponding FFT patterns of one to three rectangular areas. (**F**) HAADF-STEM image of $Pt_1Ni_1Co_1Cu_1Ru_4$ HEA containing fcc and hcp structures in a single nanoparticle. (**G**) Corresponding FFT patterns of one to two circular areas and diagram of the atomic arrangement of the fcc and hcp phases.

to the Ru-hcp HEA structure. These findings suggested that phase transformation from Pt-fcc to Ru-hcp HEA was probably controlled by overcoming energy barriers. To facilitate this transformation, a secondary heating treatment strategy was devised for the PtNiCoCuRu HEA. As outlined in Fig. 6D, the PtNiCoCuRu sample is reheated to a higher temperature, activating the atoms within the fcc structure. These activated atoms can then overcome the energy barrier, favoring the formation of the more stable Ru-hcp structure because of its lower E_{coh} during the subsequent heating phase. This approach allows for precise control over the proportions of both Ru-hcp and Pt-fcc HEA structures. To validate the efficacy of this strategy, the presentative sample of $Pt_1Ni_1Co_1Cu_1Ru_4$ was further used as an illustration. This sample was subjected to heating to 400° and 500°C in the presence of H_2. The XRD patterns are displayed in Fig. 6 (E and F). A noticeable enhancement in the (011) peak intensity of the hcp structure appears in fig. S19F, and the direct comparison is also conducted in fig. S24. Subsequently, upon heating the $Pt_1Ni_1Co_1Cu_1Ru_4$ HEA to 500°C, the characteristic peaks of the fcc structure almost entirely diminish, with the hcp structure predominantly occupying the XRD pattern in Fig. 6F. The HAADF-STEM images and corresponding FFT images in Fig. 6G further confirm the hcp structure in $Pt_1Ni_1Co_1Cu_1Ru_4$ HEA after being reheated at 500°C. This experimental demonstration strongly supported the validity of the design theory and its associated phase transition mechanism.

The post–heat treatment experiments, combined with the theoretical simulations, have substantially confirmed the potential for phase transition from fcc to hcp in $Pt_xNiCoCuRu_y$ HEA nanostructures. However, direct observation of this transition is still lacking. To address this, real-time in situ STEM experiments were conducted on the sample of $Pt_1Ni_1Co_1Cu_1Ru_4$ HEA, which was heated from

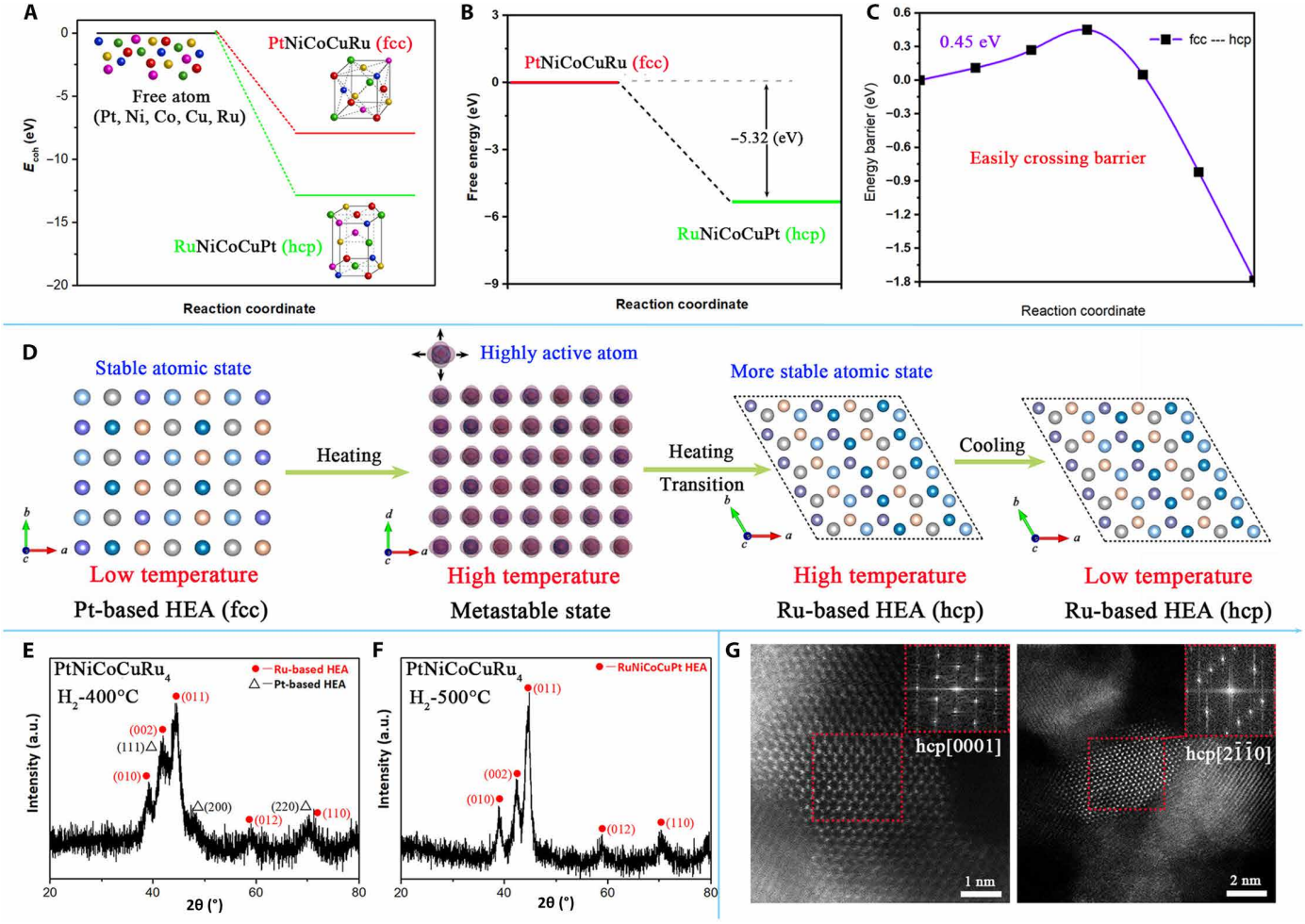


Fig. 6. Phase transition mechanism from Pt-based HEA (fcc) to Ru-based HEA (hcp). (**A**) E_{coh} of PtNiCoCuRu HEA from the free atomic state to the optimized Pt-based HEA (fcc) and Ru-based HEA (hcp) state. (**B**) ΔG of the phase transformation from the Pt-based HEA (fcc) crystal structure to the Ru-based HEA (hcp) crystal structure. (**C**) Dynamic energy barrier from Pt-based HEA (fcc) to Ru-based HEA (hcp). (**D**) Illustration of crystal structure transformation from fcc to hcp by the secondary heating process. (**E** and **F**) XRD spectra of the $Pt_1Ni_1Co_1Cu_1Ru_4$ HEA obtained at 400° and 500°C, respectively. (**G**) HAADF-STEM images of the $Pt_1Ni_1Co_1Cu_1Ru_4$ HEA-3DNFs obtained at 500°C. The inset is the corresponding FFT patterns.

room temperature to 500°C in a transmission electron microscopy chip. An area with an fcc structure was selected to initiate the in situ experiment at room temperature, as shown in Fig. 7A. No notable changes in crystal structure or morphology were observed up to 300°C (Fig. 7B). At 350°C, slight morphological changes appeared, and a small amount of the hcp structure began to emerge (Fig. 7C). By 400°C, the complete transition from the fcc structure to the hcp structure occurred, accompanied by notable morphological alterations (Fig. 7D). The structure and morphology remained stable between 400° and 450°C (Fig. 7E). At 500°C, notable morphological changes occurred because of temperature effects; however, the phase remained stable in the hcp structure (Fig. 7F). Furthermore, in situ experiments on another sample of $Pt_1Ni_1Co_1Cu_1Ru_2$ HEA (fig. S25) confirmed the phase transition from fcc to hcp at 450°C. Although reheating to 500°C caused further morphological changes, the 3DNF structure remained intact (fig. S26). Additional in situ experiments were conducted to observe phase changes of $Pt_1Ni_1Co_1Cu_1Ru_2$ during the cooling process from 500°C to room temperature. Two regions, one with an fcc structure and another with an hcp structure, were selected. No changes in either phase or morphology were observed during the cooling process (figs. S27 and S28). These in situ experiments confirmed that phase transition occurs during heating rather than the cooling process, consistent with our DFT calculations, providing direct validation of the proposed phase transition mechanism.

To thoroughly test the designed approach, a batch of PtNiCoCuRu HEA-3DNF samples with varying Pt-to-Ru atomic ratios (3:1, 2:1, 1:1, 1:2, 1:3, and 1:4) was synthesized at 300°C and subsequently reheated to 400° and 500°C in the presence of a H_2 atmosphere. The comparison of XRD spectra for these samples at different temperatures is illustrated in figs. S29 and S30. Notably, the XRD intensity for the Ru-hcp structure increased with rising temperature in each sample, supporting our phase transition predictions. The content of Ru-hcp and Pt-fcc structures was semiquantitatively calculated on the basis of the ratio of the strongest XRD peaks for hcp (011) and fcc (111), as shown in fig. S31. Thus, Fig. 7G highlights the precise control over phase composition in PtNiCoCuRu HEA samples by adjusting the reduction temperature and Pt-to-Ru atomic ratio. As

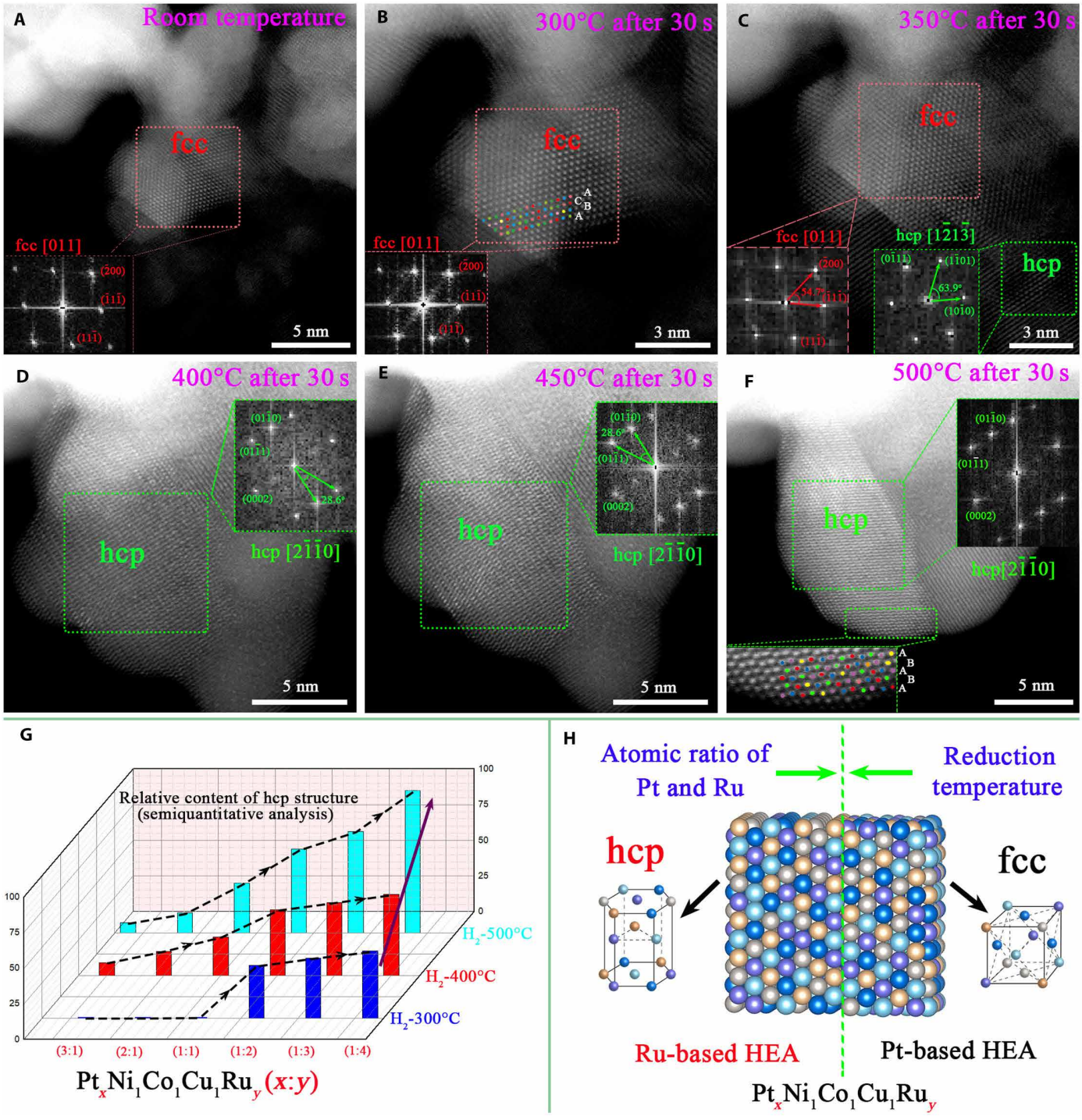


Fig. 7. Devised strategy toward the target phase structure in the PtNiCoCuRu HEA-3DNF. (**A** to **F**) In situ HAADF-STEM images and corresponding FFT patterns of the $Pt_1Ni_1Co_1Cu_1Ru_4$ HEA-3DNF at room temperature and 300°, 350°, 400°, 450°, and 500°C, respectively. (**G**) Relative content bar trends of the hcp (Ru-based HEA) structure (semiquantitative analysis) in $Pt_xNi_1Co_1Cu_1Ru_y$ HEA, showcasing different atomic ratios of Pt and Ru obtained at various temperatures. (**H**) Precise control of the target crystal structure (fcc and hcp) in the $Pt_xNi_1Co_1Cu_1Ru_y$ HEA nanoparticle by fine tuning the reduction temperature and atomic ratio of Pt and Ru.

the Ru content increases, the phase transition from the Pt-fcc structure to the Ru-hcp structure became more pronounced, enabling the coexistence of both structures within the same HEA nanoparticle. Consequently, our design strategy leverages the phase transition mechanism (Fig. 7H) to achieve the specified relative content of target phase structures (fcc and hcp) or even facilitate the concurrent presence of two-phase structures within the same PtNiCoCuRu HEA nanoparticle, thereby offering fine control over the structural composition in NM-HEA.

DISCUSSION

In summary, we have developed the SD-TDR method, a facile, cost-effective, and universal synthesis approach, to fabricate NM-HEA nanomaterials featuring a 3DNF. This approach successfully produced a range of HEA-3DNFs, including Pt-, Ir-, Ru-, and PtRu-based HEA-3DNFs, encompassing noble and transition metals (Pt, Ru, Ir, Pd, Fe, Cu, Ni, and Co). These HEA-3DNFs were synthesized under mild synthesis condition using a heating rate of 3°C/min, a thermal decomposition temperature of 300°C, and a cooling rate ranging between 5° and 10°C/min. Among these, the synthesized PtNiCoCuRuIr HEA-3DNF demonstrates exceptional MOR performance, with a mass activity of 2637 mA/mg_{Pt} (equivalent to 1054.23 mA/mg_{PtIrRu}). This performance is ~15 and 6 times higher than that of commercial Pt black (170 mA/mg_{Pt}). Furthermore, we fully investigated the critical role of NM (Pt, Ru, and Ir) in facilitating the reduction of Ni, Co, Cu, and Fe by generating highly reactive hydrogen atoms on their surfaces. This process led to the formation of single-phase solid solution alloys. We uncovered the mechanism behind the essential role of noble metals in the formation of HEA-3DNF, providing key insights into the interplay between these elements. Subsequently, we extensively explored the precise control of phase structures (fcc and hcp) within HEA-3NDF using in situ STEM experiments and DFT calculations, and the results demonstrated a phase transition from Pt-based fcc structures to Ru-based hcp structures within PtNiCoCuRu HEA. A secondary heating approach was devised to activate atoms and overcome energy barriers, facilitating this phase transformation. By manipulating the reduction temperature and atomic ratio of Pt and Ru, we achieved precise control over the relative content of fcc and hcp phases within a single HEA nanoparticle. This research establishes a fundamental understanding of NM-HEA formation mechanisms and provides a reliable approach for controlling crystal structures at the atomic level. These findings are highly promising for advancing the application of NM-HEA nanomaterials in various catalytic fields, enabling more efficient and targeted material design.

MATERIALS AND METHODS

Materials

Precious metal compounds, $(NH_4)_2PtCl_6$, $RuCl_3{\cdot}3H_2O$, $IrCl_3$, and $PdCl_2$ were purchased from Sino Platinum Metals (Yimen) Co., Ltd. (Yimen, Yunnan, P. R. China). Other compounds, $CuCl_2{\cdot}2H_2O$, $FeCl_3{\cdot}6H_2O$, $NiCl_2{\cdot}6H_2O$, $CoCl_2{\cdot}6H_2O$, NH_4Cl, and HCl (37.5 wt %) were purchased from Sigma-Aldrich (St. Louis, MO). Deionized water was used as the solvent, and all reagents and solvents were used as received without further purification.

Preparation of Pt/Ir/Ru/Pd-NiCoCuFe and $Pt_xNi_1Co_1Cu_1Ru_y$ HEA-3DNFs

The HEA-3DNFs were prepared using the spray drying technique combined with the SD-TDR method described in our previous literature (*23*, *24*). Briefly, a certain amount of precursor salts, including $(NH_4)_2PtCl_6$, $RuCl_3{\cdot}3H_2O$, $IrCl_3$, $PdCl_2$, $CuCl_2{\cdot}2H_2O$, $FeCl_3{\cdot}6H_2O$, $NiCl_2{\cdot}6H_2O$, and $CoCl_2{\cdot}6H_2O$, were mixed and dissolved in deionized water (0.001 M for each element) and heated to 60°C while being stirred ultrasonically for 2 hours. The amount of precursors was adjusted for specific purpose, with NH_4Cl consistently comprising one-fourth of the total weight of all precursor salts. The precursor solution was then spray dried using a custom spray dryer, with the following parameters: inlet temperature of 180°C, feed rate of 250 ml/hour, high-pressure gas flow rate of 1.3 liter/s, and drying gas flow rate of 35 m^3/hour. After that, the well-mixed precursors were reduced in a H_2 atmosphere, with the moderate heating rate of 3°C/min, the low thermal decomposition temperature of 300°C, the duration time of 30 min, and the cooling rate of 5° to 10°C/min. This process resulted in the successful decomposition of the precursor particles into HEA-3DNFs with the desired composition and phase structures.

Materials characterization and catalytic tests

The microstructures of samples were observed using SEM imaging (Hitachi S-8030) at an acceleration voltage of 5 kV and transmission electron microscopy (JEM-2100Plus) at an acceleration voltage of 200 kV. The elemental valence states were detected on the XPS instrument (Thermo Fisher Scientific ESCALAB Xi+) using Al Kα (1486.6 eV) radiation. XPS results were detected on an Axis Ultra DLD photoelectron spectrometer. The XRD patterns were obtained with a D8 ADVANCE (Bruker AXS, WI) using Cu-Kα radiation with a scan rate of 3°/min. The metal contents were measured by inductively coupled plasma mass spectroscopy (PerkinElmer NexION 300D ICP-MS). The atomic-scale characterization and in situ experiments were conducted on a JEM Grand ARM 300 operated at an accelerating voltage of 300 kV with both the probe and image corrector. The energy-dispersive x-ray spectra were recorded using a Dual-EDS system (JED-2300T). The performance of MOR was evaluated using a standard three-electrode electrochemical cell, comprising a Pt mesh as the working electrode, an Ag/AgCl reference electrode, and a glass carbon electrode. The catalyst ink was prepared at a concentration of 1 mg/ml, consisting of deionized water, 2-propanol, and 5% Nafion in a volume ratio of 4:1:0.025. A 10-μl aliquot of this ink was deposited onto the working electrode. The CV curves for the MOR were obtained at a scan rate of 50 mV/s, within a voltage range of −0.22 to 0.98 V (vs Ag/AgCl), in deoxygenated 0.5 M H_2SO_4 containing 0.5 M CH_3OH, following stabilization. Current stability was assessed using the CA technique, with the electrooxidation of methanol conducted at a potential of 0.65 V (vs Ag/AgCl) for 2000 s.

DFT simulations

The Vienna ab initio Simulation Package based on the DFT was used in all calculations (*43*, *44*). The plane-wave basis sets with an adequate cutoff energy of 500 eV and accurate precision was used. Electron-ion core interactions were represented by the projected augmented wave approach (*45*), while the general gradient approximation of Perdew, Burke, and Ernzerhof was used as the exchange-correlational functional to describe the interactions among electrons (*46*). A *k*-mesh grid of 4 by 4 by 1 was used for HEA supercell optimization and free

energy calculations. The convergence threshold for the energy and residual force are 10^{-5} eV and 0.01 eV/Å, respectively. The adsorption energies were calculated by using the following equation

$$E_{ads} = E_{total} - E_{sub} - E_{mol}$$

where E_{total} is the total energy of substrates and adsorbate, E_{sub} is the energy of substrates, and E_{mol} is the energy of adsorbed molecules. The free energy of each reduction step was obtained at zero bias potential using the following equation

$$\Delta G = \Delta E + \Delta E_{ZPE} + T\Delta S$$

where ΔE is the reaction energy, ΔE_{ZPE} is the difference in zero-point energies, T is the temperature (298.15 K), and ΔS is the reaction entropy.

Supplementary Materials

This PDF file includes:

Figs. S1 to S31
Tables S1 and S2

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